Research Design & Problem Identification
Essentials of a Good Research Problem
Clear and well defined
Specific/Focused (not vague)
Researchable (using available tools/data)
Manageable (neither too broad nor too narrow)
Significant/Meaningful (contributes something)
Original or offers a new angle
Sources of a Research Problem
Personal experience (day-to-day life/work)
Literature (gaps, conflicts, unanswered questions)
Discussions with peers/professionals
Social issues (community/country problems)
Theoretical research (expands academic knowledge)
New technologies and innovations
Funding agencies/government policies
Meaning & Features of Research Design
Meaning (Overall Plan/Framework)
Guides research from start to finish
Blueprint for the study
Features of a Good Design
Clearly structured (step-by-step)
Appropriate for the research problem
Flexible (room for adjustments)
Efficient (wise use of resources)
Ensures valid and reliable results
Various Steps of Research Design
Define the research problem clearly
Review available literature
Formulate the hypothesis (if applicable)
Choose the type of research (Exploratory, etc.)
Select methods for data collection
Select a sample (must represent population)
Plan for data analysis (tools/charts)
Consider ethical aspects (consent, privacy)
Write and review the research plan
Important Concepts in Research Design
Variables (things that change)
Independent variable (changed/controlled)
Dependent variable (measured/tested)
Controlled variables (kept constant)
Sampling (choosing a representative smaller group)
Validity (results truly reflect study aim)
Reliability (consistent and repeatable results)
Control (limiting outside factors/interference)
Types of Research Designs
Exploratory (little known, develops ideas, flexible)
Descriptive (answers what/where/when/how, uses surveys)
Analytical (studies relationships/causes, interprets data)
Experimental (tests cause-and-effect, changes variables)
Qualitative (focuses on thoughts/feelings, uses words)
Quantitative (based on numbers/statistics, uses measurements)
Principles of Experimental Designs
Replication (repeating experiment for consistency)
Randomisation (random participant choice to avoid bias)
Local Control (controlling all variables except the tested one)
Important Experimental Designs
Completely Randomised Design (CRD)
Randomised Block Design (RBD)
Latin Square Design (controls two factors)
Factorial Design (studies more than one factor simultaneously)